

Use-Case Flow Specification

Problem Statement #37: Group Travel Expense Simplification Engine

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SECTION: E

Use Case: UC-05 — Compute Settlement Graph

Primary Actor: Group Traveler (also usable by Trip Admin)

Includes: UC-03 Convert Currency

Preconditions:

- The trip exists with at least two Group Travelers added
- At least one expense has been logged for the trip (UC-02)
- All expense currencies have a valid daily exchange rate available.

Postconditions:

The system displays a minimal set of peer-to-peer payment suggestions such that, once all are paid, every traveler's net balance for the trip equals zero.

Main Success Scenario

1. Group Traveler selects "View Settlement" for the trip.
2. System retrieves all logged expenses and their participant splits for the trip.
3. System invokes UC-03 (Convert Currency) to convert every foreign-currency expense into the trip's base currency using that expense's date rate.
4. System computes each traveler's net balance (total paid minus total owed) in the base currency.
5. System runs the min-flow settlement algorithm to generate the smallest set of payment edges that brings every balance to zero.
6. System displays the settlement graph as a list of suggested payments (payer, payee, amount).
7. Use case ends successfully.

Alternate Flow

4a. Missing Exchange Rate

- 4a1. System detects that no exchange rate exists for an expense's currency and date.
- 4a2. System flags the affected expense and excludes it from the balance calculation, displaying a warning banner: "1 expense could not be converted — rate unavailable."
- 4a3. System proceeds to compute the settlement graph using the remaining convertible expenses.
- 4a4. Trip Admin is notified to manually enter or retry the missing exchange rate; once entered, the traveler can re-run step 1 to include it.